Part A · Concept Notes

1. Biological Databases & Data Retrieval

NCBI — U.S. National Library of Medicine's biological database hub. Hosts GenBank (nucleotides), Gene, Protein, Taxonomy, and PubMed. Used to pull raw sequence records and literature.

UniProt — Central protein database. **Swiss-Prot** = manually reviewed / high confidence; **TrEMBL** = automatic / unreviewed. Provides function, domains, PTMs, and disease links.

PDB (Protein Data Bank) — Global repository of 3D macromolecular structures solved by X-ray crystallography, NMR, or Cryo-EM. Used for structural analysis and docking studies.

2. Sequence Similarity Search — BLAST

Compares a query sequence against a database to find homologous sequences.

- **BLASTn** — nucleotide vs nucleotide
- **BLASTp** — protein vs protein
- **BLASTx** — translated nucleotide vs protein
- **tBLASTn** — protein vs translated nucleotide
- **tBLASTx** — translated nucleotide vs translated nucleotide

Key metrics: **E-value** (lower = more significant), **% identity**, **bit score**, **query coverage**.

3. Multiple Sequence Alignment (MSA)

ClustalW — progressive alignment algorithm; builds a guide tree, then aligns most-related sequences first. Command-line based (ClustalX = GUI version).

MEGA — integrated GUI suite combining MSA, model selection, and tree-building in one package. Popular in teaching and research for its accessibility.

4. Phylogenetic Analysis

Trees can be generated directly from ClustalW's alignment (Neighbor-Joining) or built more rigorously in MEGA using:

- **NJ** — Neighbor-Joining
- **ML** — Maximum Likelihood
- **MP** — Maximum Parsimony
- **UPGMA**

Bootstrap analysis (typically 500–1000 replicates) tests the statistical confidence of each branch.

5. Tree Visualization — iTOL

Web-based tool for displaying, annotating, and refining phylogenetic trees (input: Newick / Nexus / phyloXML). Supports circular/rectangular layouts, color strips, metadata overlays, and vector export (SVG/PDF/EPS) for publication-quality figures.

End-to-end workflow: Retrieve (NCBI/UniProt/PDB) → Search homologs (BLAST) → Align (ClustalW/MEGA) → Build tree (ClustalW/MEGA + bootstrap) → Visualize & refine (iTOL).

Part C · File Name Format & Description Reference

Exact files to generate and upload to GitHub for each stage, with format and purpose.

1. Data Retrieval

NCBI — Nucleotide Sequence

File	Format	Description
GenBank record	`.gb`	Full annotated record — organism, features, references, sequence
Sequence file	`.fasta`	Raw nucleotide sequence with header
Description	`.txt`	2–3 line note on gene/organism and reason for selection

NCBI — Protein Sequence

File	Format	Description
GenPept record	`.gp`	Annotated protein record with CDS translation info
Sequence file	`.fasta`	Raw protein sequence
Description	`.txt`	Brief note on protein identity/function

PDB

File	Format	Description
Sequence file	`.fasta`	Amino acid sequence extracted from the structure
Legacy structure file	`.pdb`	Atomic coordinates in Legacy PDB format
Validation report	`.pdf`	Full structure-quality report (resolution, geometry, clash score)

UniProt

File	Format	Description
Entry file	`.xml`	Structured record — function, features, cross-references
Entry file	`.txt`	Flat-file version of the same entry
Sequence file	`.fasta`	Raw protein sequence

2. BLAST (repeat set for every BLAST type used — min. 20 hit sequences each)

File	Format	Description
Query sequence	`.fasta`	Input sequence used to run BLAST
Alignment/hit table	`.csv`	Accession, description, % identity, E-value, bit score, coverage
Hit sequences (complete)	`.fasta`	Combined FASTA of ≥20 matched/retrieved sequences

Apply to each: BLASTn, BLASTp, BLASTx, tBLASTn, tBLASTx (as required) — keep each in its own labeled folder.

3. MSA — ClustalW

File	Format	Description
Alignment file	`.aln`	ClustalW multiple sequence alignment output
Combined input file	`.fasta`	All sequences submitted together for alignment
Tree file(s)	`.dnd` / `.nwk`	Every tree file generated by ClustalW (guide tree + outputs)

4. MSA & Phylogenetics — MEGA

File	Format	Description
Sequence dump file	`.fasta`	Combined sequences exported/imported into MEGA
Alignment session file	`.mas`	MEGA's native alignment session file
Aligned sequence file	`.meg` / `.fasta`	Final aligned sequences used for tree building
Newick tree file	`.nwk`	Final phylogenetic tree
Tree images	`.png`	Every generated tree view (NJ, ML, MP, bootstrap consensus, etc.)

Suggested GitHub Folder Structure

`01_Data_Retrieval/` → NCBI_Nucleotide, NCBI_Protein, PDB, UniProt
`02_BLAST/` → BLASTn, BLASTp, BLASTx (one folder per type)
`03_MSA_ClustalW/` → alignment.aln, combined_input.fasta, tree_files/
`04_MEGA/` → sequence_dump.fasta, alignment.mas, aligned_sequences.meg, tree.nwk, tree_images/
`README.md` → overview + short description of workflow & each file